

COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No. 04

Combinational Circuit

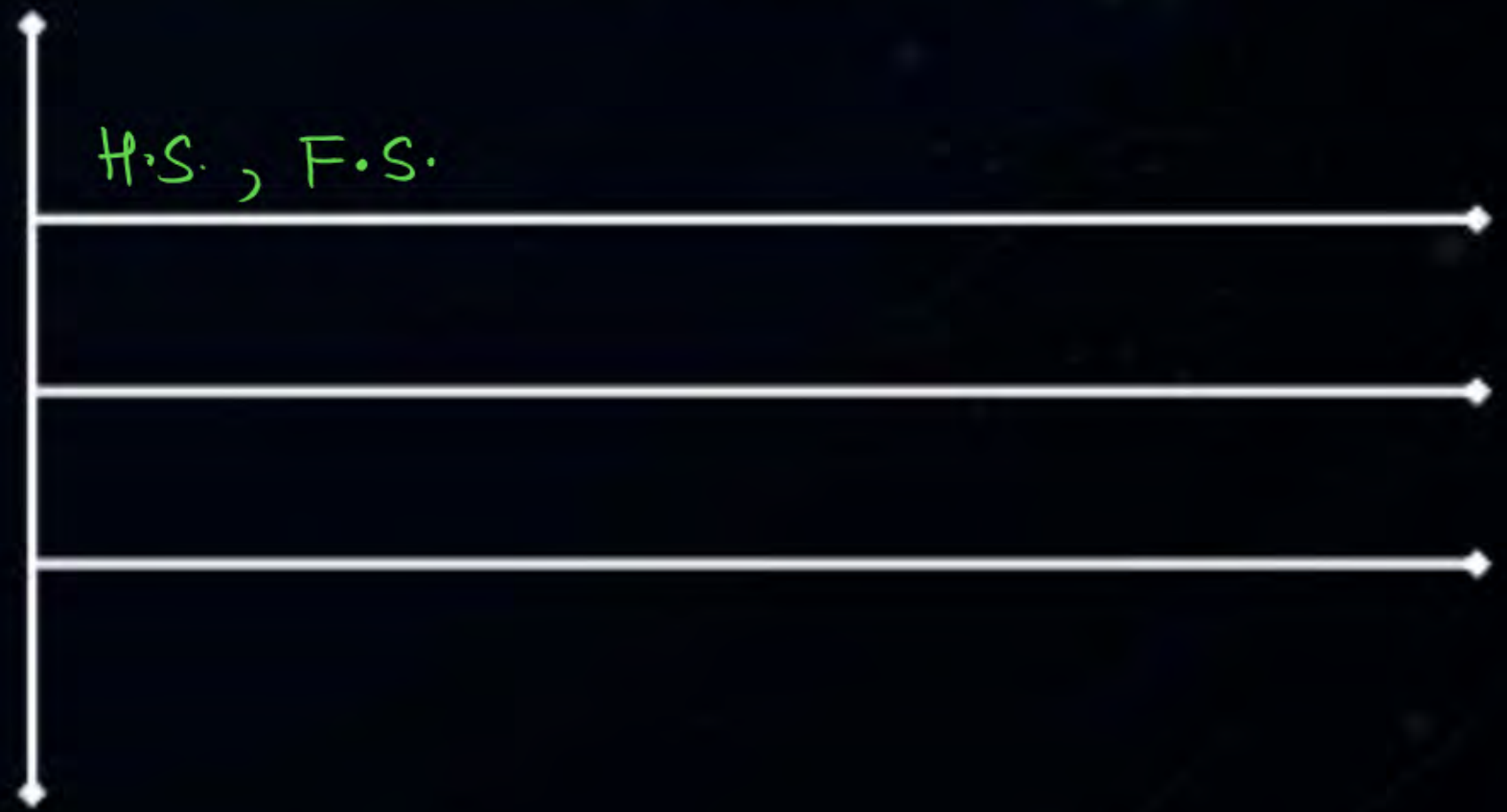


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Recap of Previous Lecture

H.S. , F.S.





Topics to be Covered

K-Map



[K-Map]

Used for Entries :

- '1'
- '0'
- 'X' $\longrightarrow$ don't care
- 'Variables'

$$f(A, B, C) = \sum(1, 2, 3, 6, 7)$$

		$\overline{B}\overline{C}$	$\overline{B}C$	BC	$B\overline{C}$
		00	01	11	10
$\overline{A}$	0		1	1	1
A	1			1	1

$\overline{A}C$

$$\begin{aligned}\overline{A}\overline{B}C + \overline{A}BC \\ \overline{A}C[\overline{B} + B] &= \overline{A}C\end{aligned}$$

- Why gray code is used in case of K-map ?

	$\overline{B}\overline{C}$	$\overline{B}C$	$B\overline{C}$	BC
$\overline{A}$		1	1	
A				

	$\overline{B}\overline{C}$	$\overline{B}C$	BC	$B\overline{C}$
$\overline{A}$		1	1	
A	4	5	7	6

$$\overline{A}\overline{B}C + \overline{A}B\overline{C}$$

$$\overline{A}[\overline{B}C + B\overline{C}]$$

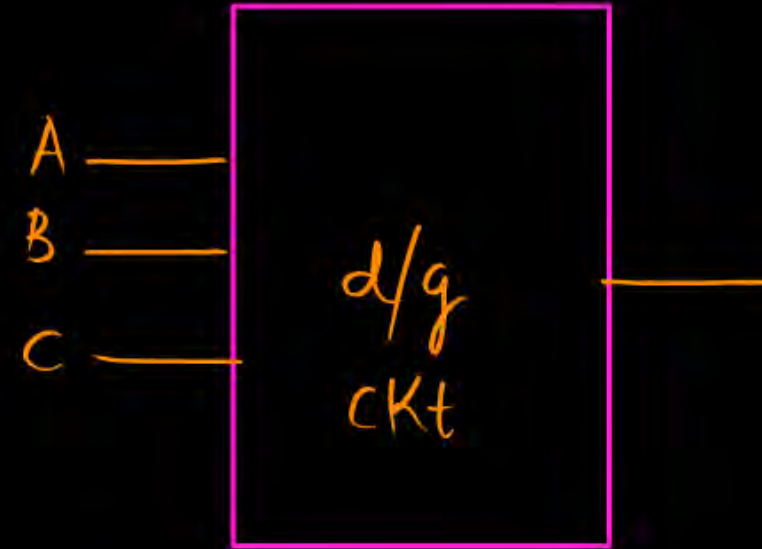
$$\overline{A}C$$

$$\overline{A}\overline{B}C + \overline{A}BC$$

$$AC[\overline{B} + B]$$

Binary	gray
$(24)_{10} (11000)_2$	$(10100)_{\text{gray}}$
$(25)_{10} (11001)_2$	$(10101)_{\text{gray}}$
$(26)_{10} (11010)_2$	$(10111)_{\text{gray}}$
$(27)_{10} (11011)_2$	$(10110)_{\text{gray}}$
$(28)_{10} (11100)_2$	$(10010)_{\text{gray}}$

- What is the meaning of don't care X?



$$B=0 \text{ \& } C=1$$

	B	C
0	0	1
1	0	1

BCD code (8421 Code)

0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001

1010
1011
1100
1101
1110
1111

Invalid
BCD
Code

$$(24)_{10} = (11000)_2 \\ = (\boxed{0010}\boxed{0100})_{BCD}$$



#Q. Design a digital ckt which takes BCD code as i/p and generates o/p '1' if it is divisible by '3'.

• 4 i/p lines P, Q, R, S

O/p line $\rightarrow$ Y

$Y[P, Q, R, S]$

$$= \sum(0, 3, 6, 9)$$

$$+ d\sum(10, 11, 12, 13, 14, 15)$$

P	Q	R	S	Y
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	X
1	0	1	1	X
1	1	0	0	X
1	1	0	1	X
1	1	1	0	X
1	1	1	1	X

[2-variable, 3-variable and 4-variable K-maps] $f(A, B, C, D)$



$f(A, B)$

	$\overline{B}$	B
$\overline{A}$	0	1
A	2	3

$f(A, B, C)$

	$\overline{B}\overline{C}$	$\overline{B}C$	BC	$B\overline{C}$
$\overline{A}$	0	1	3	2
A	4	5	7	6

	$\overline{A}$	A
$\overline{B}\overline{C}$	0	4
$\overline{B}C$	1	5
BC	3	7
$B\overline{C}$	2	6

	$\overline{C}\overline{D}$	$\overline{C}D$	CD	$C\overline{D}$
$\overline{A}\overline{B}$	0	1	3	2
$\overline{A}B$	4	5	7	6
AB	12	13	15	14
$A\overline{B}$	8	9	11	10

[Question]



$$f(A, B) = \Sigma(1, 3)$$

	$\overline{B}$	B
$\overline{A}$		1
A		1

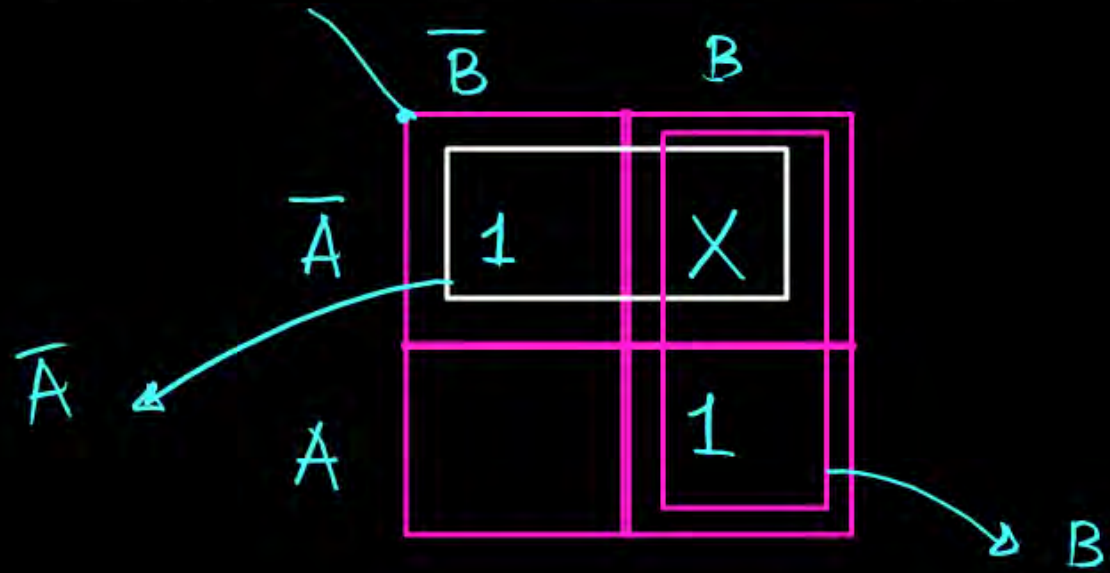
An arrow points from the '1' in the cell where $A=1$ and $B=1$ to the label B .

$$f(A, B) = B$$

[Question]



$$f(A, B) = \Sigma(0, 3) + d\Sigma(1)$$

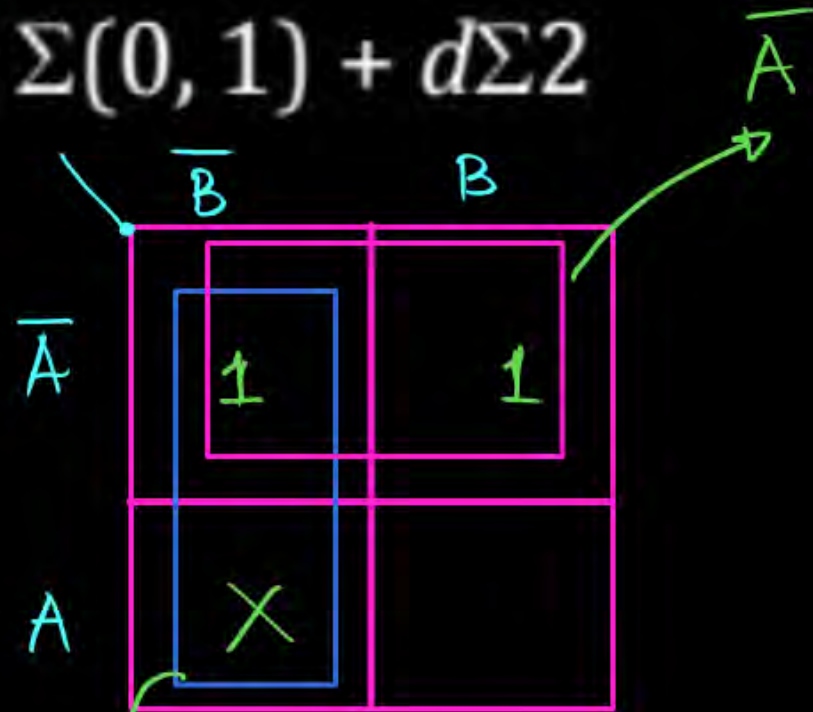


$$f(A, B) = \overline{A} + B$$

[Question]

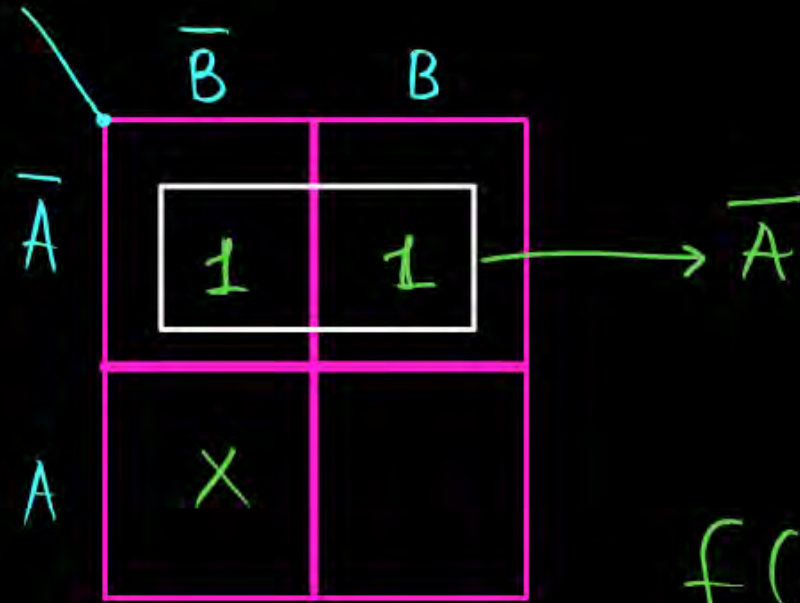


$$f(A, B) = \Sigma(0, 1) + d\Sigma 2$$



$\bar{B}$ (redundant)

$$f(A, B) = \bar{A}$$



$$f(A, B) = \bar{A}$$

[Question]



$$f(A, B, C) = \Sigma(0, 1, 5, 7)$$

	$\overline{B}\overline{C}$	$\overline{B}C$	BC	$B\overline{C}$
$\overline{A}$	1	1		
A		1	1	

$\overline{A}\overline{B}$ (from $\overline{A}$ row, $\overline{B}\overline{C}$ column)
 $\overline{B}C$ (from $\overline{B}C$ column)
 AC (from A row, BC column)

$$\begin{aligned}
 &= \overline{A}\overline{B} + AC + \overline{B}C \\
 &\quad \text{redundant} \\
 &= \overline{A}\overline{B} + AC
 \end{aligned}$$

	$\overline{B}\overline{C}$	$\overline{B}C$	BC	$B\overline{C}$
$\overline{A}$	1	1		
A		1	1	

$\overline{A}\overline{B}$ (from $\overline{A}$ row, $\overline{B}\overline{C}$ column)
 AC (from A row, BC column)
 redundant group (from $\overline{B}C$ column)

$$= \overline{A}\overline{B} + AC$$

[Question]

$$f(A, B, C) = \Sigma(0, 1, 3, 6, 7)$$

	$\overline{B}\overline{C}$	$\overline{B}C$	BC	$B\overline{C}$
$\overline{A}$	1	1	1	
A			1	1

$$= \overline{A}\overline{B} + \overline{A}C + AB$$

or

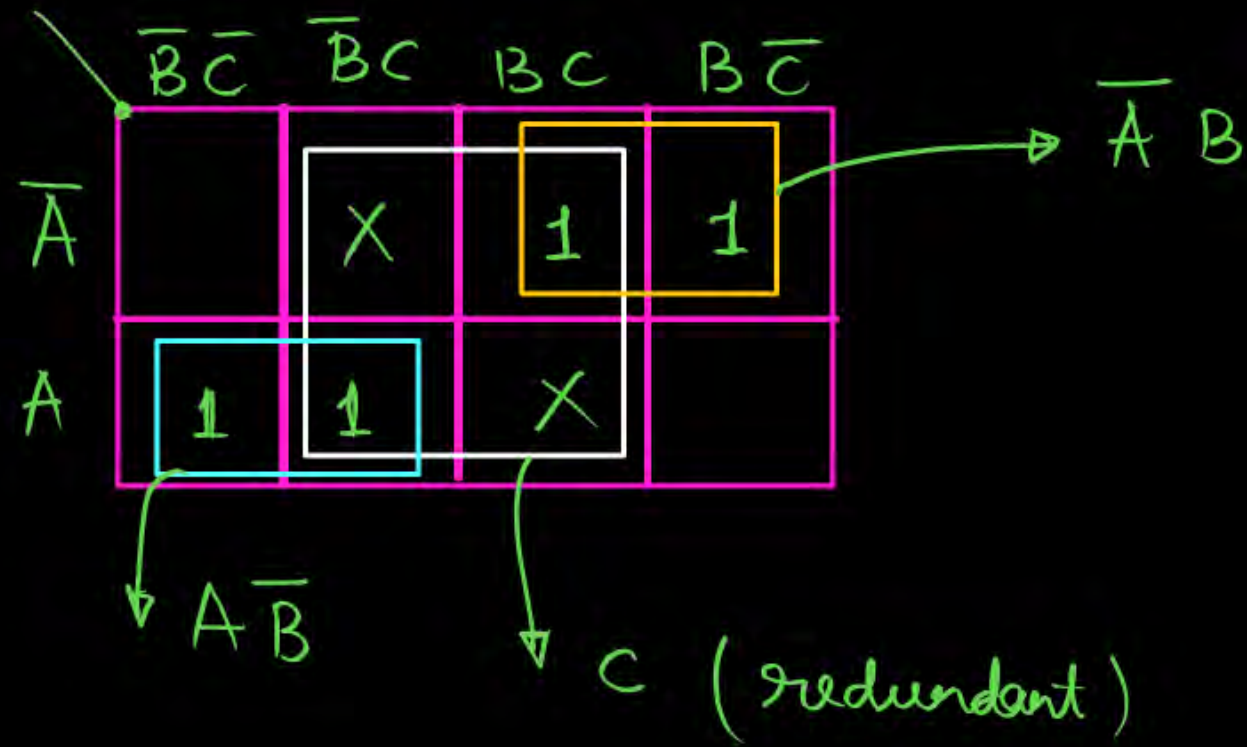
	$\overline{B}\overline{C}$	$\overline{B}C$	BC	$B\overline{C}$
$\overline{A}$	1	1	1	
A			1	1

$$\overline{A}\overline{B} + AB + BC$$

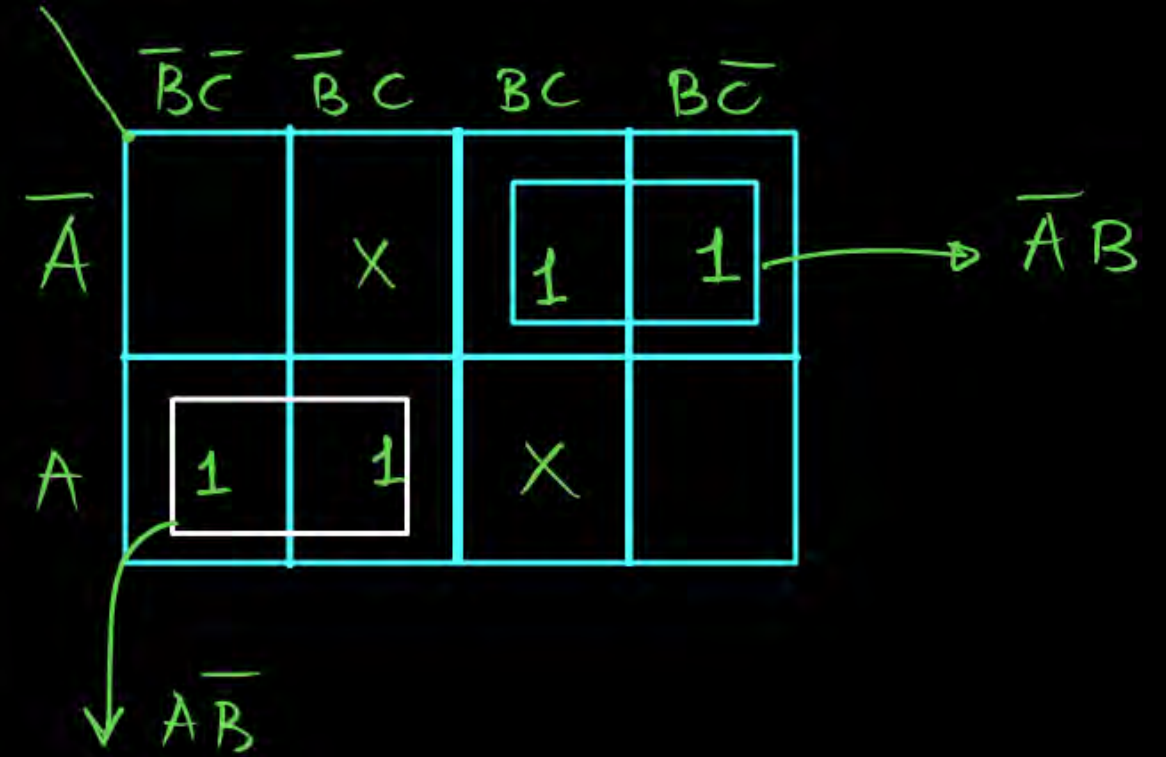
Note: K-Map may have more than one solution but it is not always necessary.

[Question]

$$f(A, B, C) = \Sigma(2, 3, 4, 5) + d\Sigma(1, 7)$$

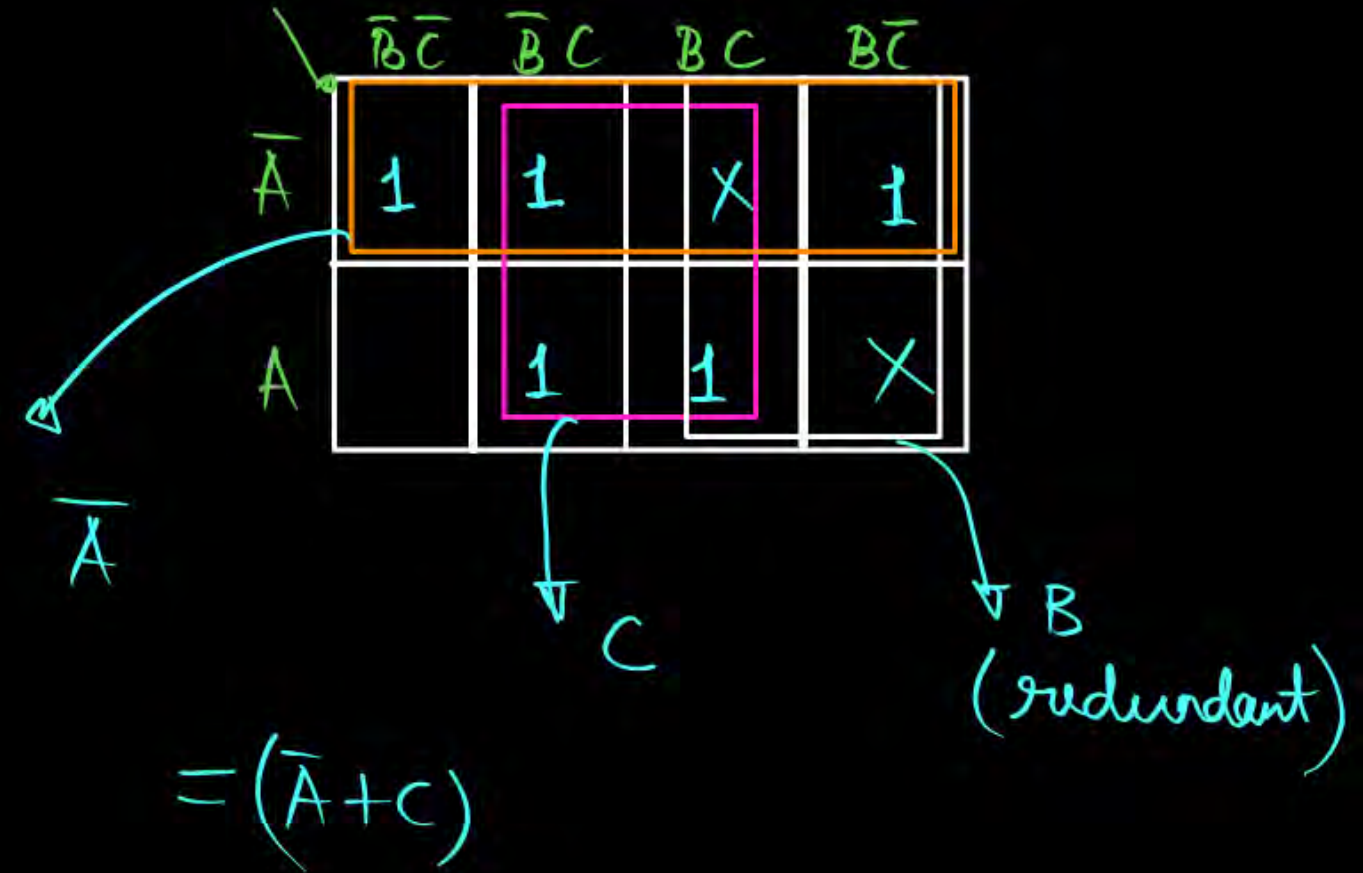
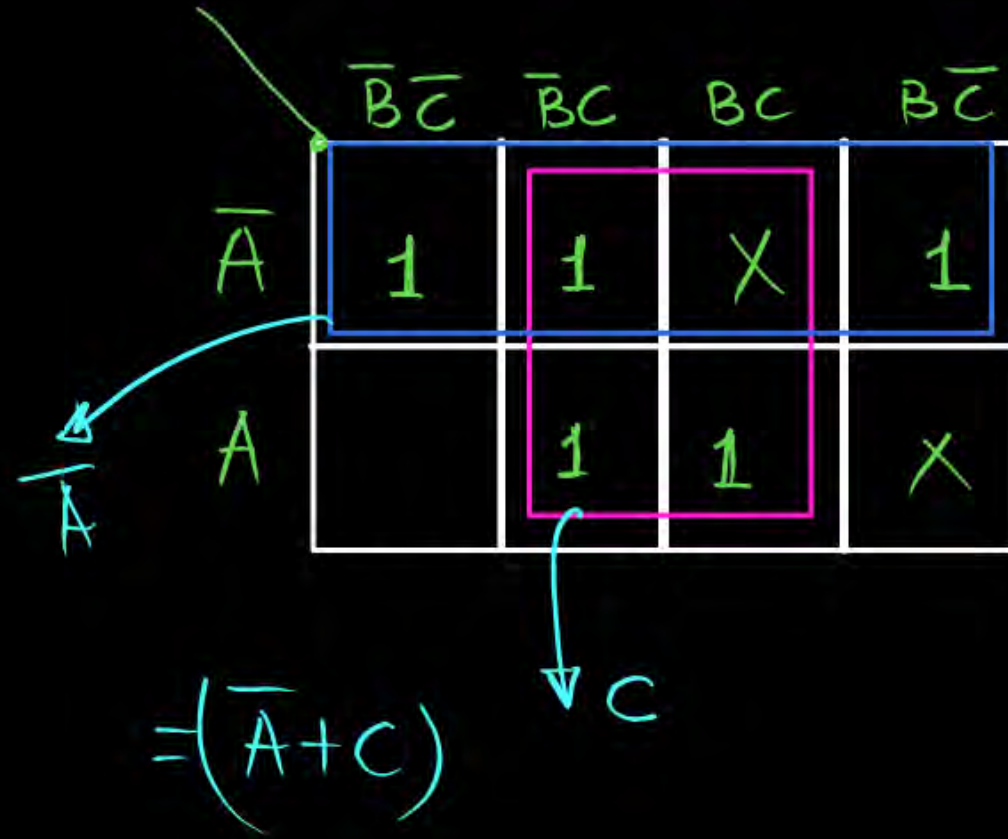
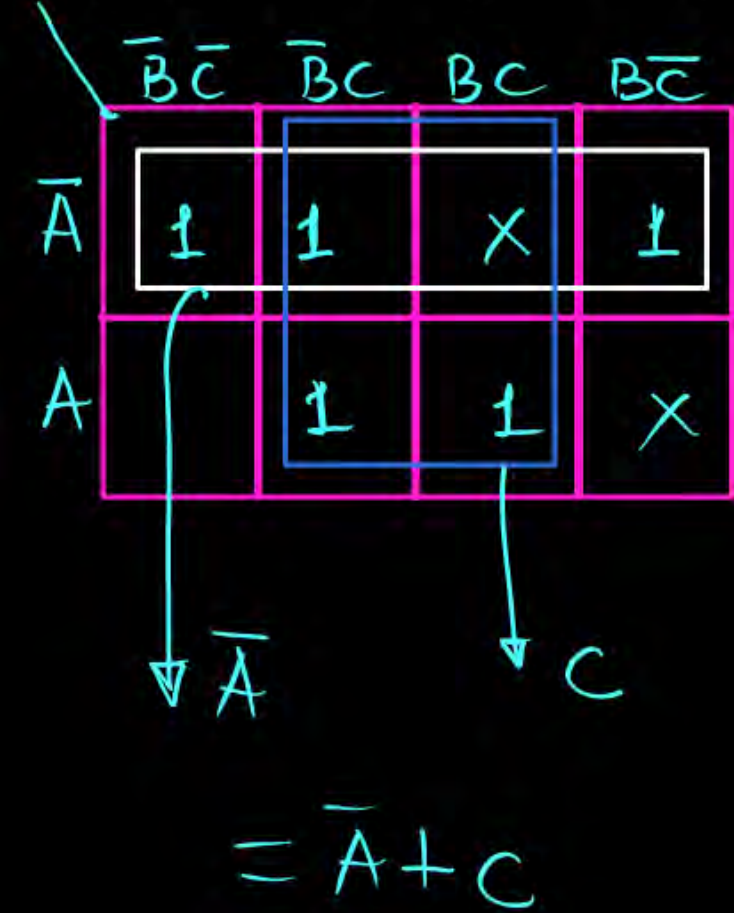


$$= \overline{A}B + \overline{A}\overline{B}$$



[Question]

$$f(A, B, C) = \Sigma(0, 1, 2, 5, 7) + d\Sigma(3, 6)$$



Thank you

GW
Soldiers !

